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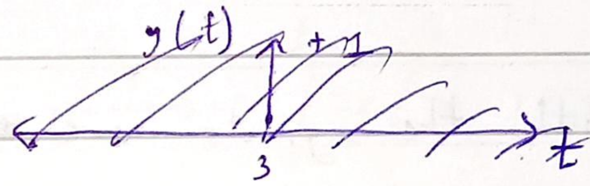
ECS.101 - Assignment 2

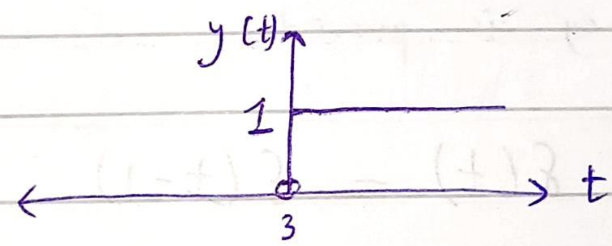
1\. a) $x(t) = \delta(t)$

$\Rightarrow \Delta_3(x(t)) = \delta(t-3) = z(t)$

$\Rightarrow A(z(t)) = \int_{-\infty}^{t} \delta(\tau-3)\, d\tau = y(t)$

$$\Rightarrow y(t) = \begin{cases} 0, & t < 3 \\ 1, & t \geq 3 \end{cases}$$





b) Yes, the system is linear. Let the sys. operator be H.

Let : $x_1(t) \xrightarrow{H} y_1(t)$

$x_2(t) \xrightarrow{H} y_2(t)$

$\Rightarrow y_1(t) = A(\Delta_3(x_1(t))) = \int_{-\infty}^{t} x_1(\tau-3)\, d\tau$

$\Rightarrow y_2(t) = A(\Delta_3(x_2(t))) = \int_{-\infty}^{t} x_2(\tau-3)\, d\tau$

Now, $\alpha x_1(t) + \beta x_2(t) \xrightarrow{H} A(\Delta_3(\alpha x_1(t) + \beta x_2(t)))$

$= \int_{-\infty}^{t} (\alpha x_1(\tau-3) + \beta x_2(\tau-3))\, d\tau$

$= \underbrace{\alpha \int_{-\infty}^{t} x_1(\tau-3)\, d\tau}_{y_1(t)} + \underbrace{\beta \int_{-\infty}^{t} x_2(\tau-3)\, d\tau}_{y_2(t)}$

$\therefore \alpha x_1(t) + \beta x_2(t) \xrightarrow{H} \alpha y_1(t) + \beta y_2(t)$

$\therefore$ H.P.

c) Yes, the system is time-invariant.

- $x(t) \longrightarrow \boxed{\Delta} \longrightarrow x(t-\Delta) \xrightarrow{H} \int_{-\infty}^{t} x(\tau-\Delta-3)\,d\tau$

- $x(t) \xrightarrow{H} \int_{-\infty}^{t} x(\tau-3)\,d\tau \longrightarrow \boxed{\Delta} \longrightarrow \int_{-\infty}^{t-\Delta} x(\tau-3)\,d\tau$

$= \int_{-\infty}^{t} x(\tau-\Delta-3)\,d\tau$ (u-substitution and changing limits)

Since both the signals are equivalent, time-invariancy is proved.

d) i) $x(t) = \delta(t) - \delta(t-1)$

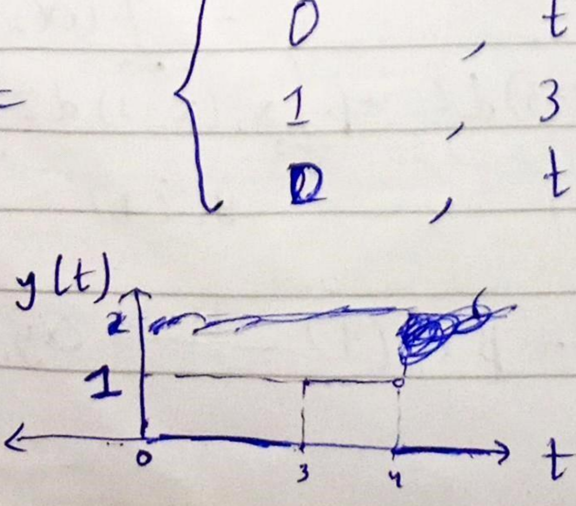
$\delta(t) \xrightarrow{H} \int_{-\infty}^{t} \delta(\tau-3)\,d\tau$ — ①

$\delta(t-1) \xrightarrow{H} \int_{-\infty}^{t} \delta(\tau-1-3)\,d\tau$ — ②

Since H is linear, $\delta(t) - \delta(t-1) \xrightarrow{H}$ ① − ②

$= \begin{cases} 0, & t<3 \\ 1, & t \geq 3 \end{cases} - \begin{cases} 0, & t<4 \\ 1, & t \geq 4 \end{cases}$

$= y(t) = \begin{cases} 0, & t<3 \\ 1, & 3 \leq t < 4 \\ 0, & t \geq 4 \end{cases}$



ii) $x(t) = u(t) + u(t-1)$

$$u(t) \xrightarrow{H} \int_{-\infty}^{t} u(\tau-3)\,d\tau$$

$$= \begin{cases} \int_{-\infty}^{3} 0\,d\tau + \int_{3}^{t} 1\,d\tau & (t>3) \\ 0 & (t\le 3) \end{cases}$$

$$= \begin{cases} t-3, & t>3 \\ 0, & t\le 3 \end{cases} \quad - (3)$$

$$u(t-1) \xrightarrow{H} \int_{-\infty}^{t} u(\tau-1-3)\,d\tau$$

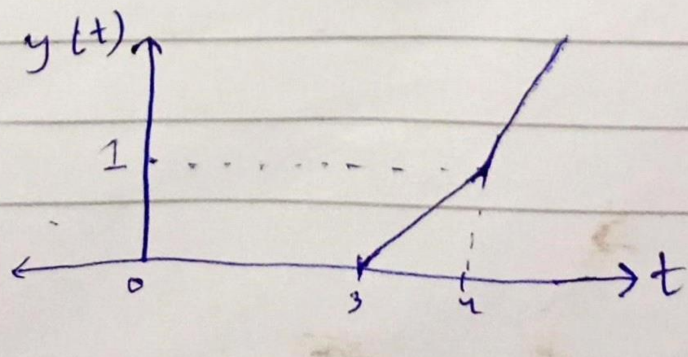
$$= \begin{cases} \int_{-\infty}^{4} 0\,d\tau + \int_{4}^{t} 1\,d\tau & (t>4) \\ 0 & (t\le 4) \end{cases}$$

$$= \begin{cases} t-4, & t>4 \\ 0, & t\le 4 \end{cases} \quad - (4)$$

Since H is linear, $u(t)+u(t-1) \xrightarrow{H} (3)+(4)$

$$= \begin{cases} t-3, & t>3 \\ 0, & t\le 3 \end{cases} + \begin{cases} t-4, & t>4 \\ 0, & t\le 4 \end{cases}$$

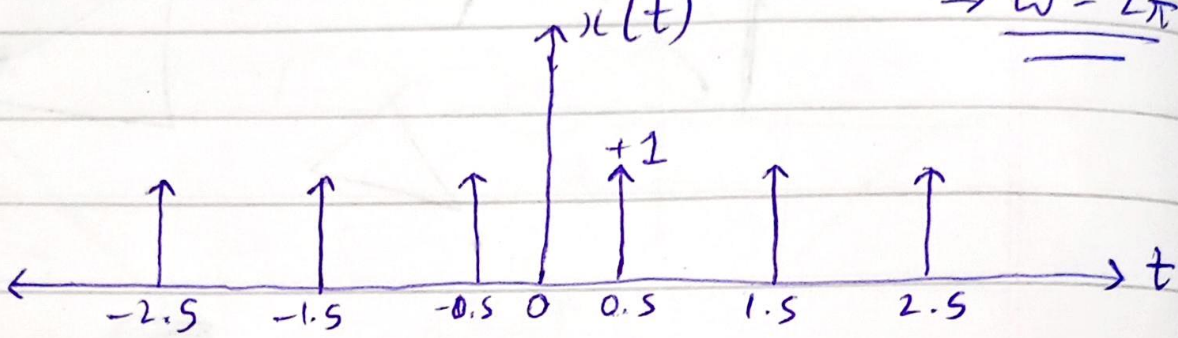
$$= \begin{cases} 0, & t\le 3 \\ t-3, & 3<t\le 4 \\ 2t-7, & t>4 \end{cases} = y(t)$$



Q2. $x(t) = \delta(t-0.5)$, $0 \le t \le 1$, $T = 1$

$\Rightarrow \omega = 2\pi$

a)



b) $x(t) \xleftrightarrow{FS} \{a_k\}$

$$\Rightarrow a_k = \frac{1}{T}\int_{\langle T\rangle} x(t)\, e^{-jk\omega t}\, dt$$

$$= \int_0^1 x(t)\, e^{-jk2\pi t}\, dt = \int_0^1 \delta(t-0.5)\, e^{-jk2\pi t}\, dt$$

$$= e^{-jk2\pi(0.5)} \int_0^1 \delta(t-0.5)\, dt \quad \text{[Sifting Property]}$$

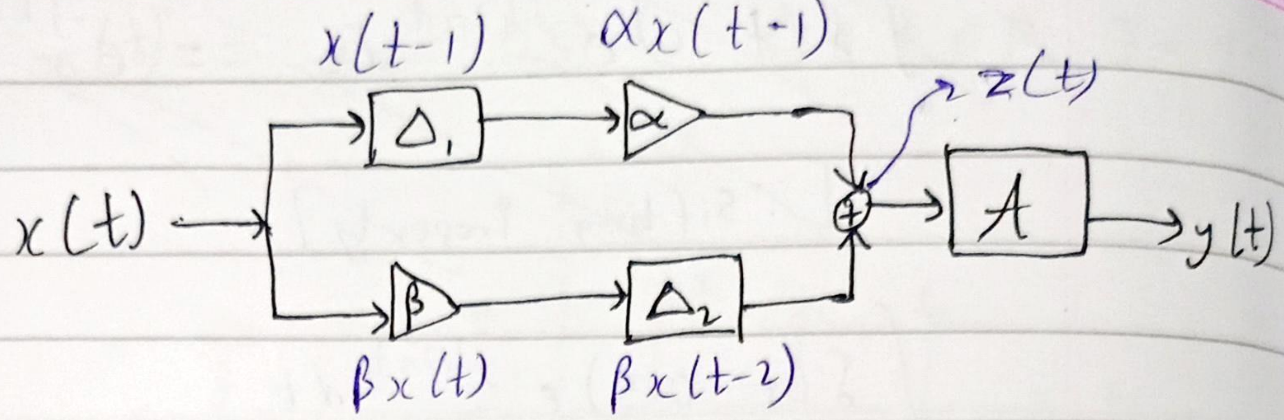
$$= e^{-jk\pi}\cdot(1) = e^{-jk\pi}$$

$$\Rightarrow a_k = \begin{cases} 1 & , k = 2m \\ -1 & , k = 2m-1 \end{cases} \quad ; m \in \mathbb{Z}$$

$$\therefore x(t) = \ldots + e^{-2j\omega t} - e^{-j\omega t} + 1 - e^{j\omega t} + e^{2j\omega t} - \ldots$$

$$= \delta(t-0.5)$$

Q3. a)



$$z(t) = \alpha x(t-1) + \beta x(t-2)$$

$$\Rightarrow A(z(t)) = \int_{-\infty}^{t} [\alpha x(\tau-1) + \beta x(\tau-2)]\, d\tau$$

$$\therefore y(t) = \alpha \int_{-\infty}^{t} x(\tau-1)\, d\tau + \beta \int_{-\infty}^{t} x(\tau-2)\, d\tau$$

~~b) $x(t) \rightarrow \boxed{\Delta_1} \rightarrow \alpha \rightarrow x$~~

b) $x(t) \rightarrow \boxed{\Delta_1} \rightarrow x(t-1) \rightarrow \boxed{\alpha} \rightarrow \alpha x(t-1)$

$x(t) \rightarrow \boxed{\alpha} \rightarrow \alpha x(t) \rightarrow \boxed{\Delta_1} \rightarrow \alpha x(t-1)$ ≡

$x(t) \rightarrow \boxed{\Delta_2} \rightarrow x(t-2) \rightarrow \boxed{\beta} \rightarrow \beta x(t-2)$

$x(t) \rightarrow \boxed{\beta} \rightarrow \beta x(t) \rightarrow \boxed{\Delta_2} \rightarrow \beta x(t-2)$ ≡

∴ The output signal is same regardless of the order of delay & scaling blocks.

(The adder output, $z(t)$, gives the same signal through A regardless of delay-scaling order)

Q4. $$y(t) = \int_{-\infty}^{t} x^2(r)\,dr \quad , \quad \forall\, t \in \mathbb{R}$$

a) • $$y_1(t) = \int_{-\infty}^{t} x_1^2(r)\,dr$$

$$y_2(t) = \int_{-\infty}^{t} x_2^2(r)\,dr$$

$$\int_{-\infty}^{t} [\alpha x_1(r) + \beta x_2(r)]^2\,dr \neq \alpha y_1(t) + \beta y_2(t)$$

∴ The system is $\boxed{\text{non-linear}}$.

• $$x(t) \longrightarrow \boxed{\Delta} \longrightarrow x(t-\Delta) \xrightarrow{H} \int_{-\infty}^{t} x^2(r-\Delta)\,dr$$

$$\equiv$$

$$x(t) \xrightarrow{H} \int_{-\infty}^{t} x^2(r)\,dr \longrightarrow \boxed{\Delta} \longrightarrow \int_{-\infty}^{t-\Delta} x^2(r)\,dr$$

∴ The system is $\boxed{\text{time-invariant}}$. [u-substitution & limits change]

• The system only requires input till the current times, as it integrates till 't'.

∴ The system is $\boxed{\text{causal}}$.

b) $$x(t) = u(t) = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

$$\Rightarrow y(t) = \int_{-\infty}^{t} x^2(r)\,dr = \int_{-\infty}^{t} u^2(r)\,dr$$

$$\Rightarrow y(t) = \begin{cases} \overset{0}{\cancel{\int_{-\infty}^{0} u^2(r)\,dr}} + \int_0^t \underbrace{u^2(r)}_{1}\,dr & (t \geq 0) \\ 0 & (t < 0) \end{cases}$$

$$\Rightarrow y(t) = \begin{cases} t & , \ t \geq 0 \\ 0 & , \ t < 0 \end{cases}$$